Centenary Celebrated Sharnbasveshwar Vidya Vardhaka Sangha's



ಶರಣಬಸವ Sharnbasva ವಿಶ್ವವಿದ್ಯಾಲಯ University

ಕಲಬುರಗಿ–585 103
ಕರ್ನಾಟಕ–ಭಾರತ
ಸ್ಥಾಪನೆ : 2017

Kalaburagi-585 103
Karnataka-India
Estd.: 2017



Faculty of: Engineering and Technology	Department: CSE/CSD Program: B.Tech	AY: 2026-27

INTERNAL ASSESSMENT (I.A.) TEST: I

Course Title: Operating Systems	Course Code: 25CS304
Date: 04/11/2026	Time: 2.00 pm to 3.30 pm
Semester and Section: III Sem	Max. Marks:30

Note: Answer any three full questions selecting at least one question from each module

Q. No.		QUESTIONS	Marks	CO's	RBT Levels
		MODULE-I			
1	a)	Identify the different types of system calls and relate each type to appropriate operating-system operations.	06	CO1	L3
	b)	Interview the abstract view of a computer system with a neat diagram and show how its major components interact with the operating system.	06	CO1	L3
		OR			
2	a)	Utilize the mechanisms of inter-process communication and compare shared-memory and message-passing approaches.	06	CO1	L3
	b)	Identify the process concept and explain how process states and process operations are related during process execution	06	CO1	L3
		MODULE-II			
3	a)	Compare the many-to-one, one-to-one, and many-to-many multithreading models with respect to their execution characteristics.	06	CO2	L4
	b)	Consider the following snapshot of a system **Process / Arrival Time / Burst time** P0 / 0 / 3 P1 / 2 / 6 P2 / 4 / 4 P3 / 6 / 5 P4 / 8 / 2 Consider the following set of processes with CPU burst time(in ms). Compute the waiting time and average turnaround time forthe above process using FCFS, SJF non preemptive scheduling algorithm.	06	CO2	L3
		OR			
4	a)	Examine the benefits and challenges of multithreaded programming in multi-core systems with suitable examples.	06	CO2	L4

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	b)	Demonstrate the handling of threading issues such as fork(), exec(), signal handling, thread cancellation, and thread-local storage using suitable examples.	06	CO2	L3
		MODULE-III			
5	a)	Explain the three requirements that a solution to the critical-section problem must satisfy.	06	CO3	L2
		OR			
6	a)	Outline and explain the structure of the writer process and reader process.	06	CO3	L2

Marks Distribution of CO's and RBT Level wise

CO's	L1	L2	L3	L4	Total Marks	% of Marks Allotted
CO1	-	-	24	-	24	40%
CO2	-	-	12	12	24	40%
CO3	-	12		-	12	20%
Total RBT Marks	-	12	36	12	60	100%
% of RBT Levels	-	20%	60%	20%-	100%	100%